CoopGCN: Axiomatic credit assignment in graph convolutional networks via tri-level cooperative game theory for robust, preference-aware recommendation

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Abstract

Linearised graph convolutional networks (GCNs), most notably LightGCN, have become the default backbone for collaborative filtering by discarding feature transformations and nonlinear activations while retaining neighbourhood aggregation. We argue that this simplification removed something the literature has not replaced: every mechanism for *differential credit assignment*. In LightGCN the influence of a neighbour is fixed at $1/\sqrt{(d_u\ d_i)}$, a purely topological constant, so an incidental misclick and a deliberate niche purchase propagate with identical relative importance. We first give a systematic taxonomy of twelve representational and training-level weaknesses of linear graph CF, and show that the most severe of them share three root causes: credit is flat, structure is pairwise-only, and the training recipe amplifies the head of the popularity distribution. We then observe that neighbourhood aggregation is formally a *cooperative credit-assignment game*, and introduce **CoopGCN**, which instantiates this view at three structural levels: an edge-level game ($\mathbf{G_1}$) modulating pairwise message weights by Monte-Carlo Shapley values, a hyperedge-level game ($\mathbf{G_2}$) weighting beyond-pairwise coalitions, and a data-level game ($\mathbf{G_3}$) applying truncated Monte-Carlo Shapley valuation to reweight and prune training interactions. Because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy player and additivity, credit assignment becomes axiomatic rather than heuristic: we prove that LightGCN and degree-scaled norm weighting are special cases of the framework, and that the influence of an uninformative edge is bounded by O(ε) rather than $\Theta(1/d_u\ d_{i\wedge*})$. A stop-gradient consistency loss L_{game} distils exponential moving-average Shapley credits into learnable attention, confining all game-theoretic computation to training and leaving inference cost unchanged. Under a leakage-audited temporal protocol with full-catalogue ranking at cut-off 20, CoopGCN attains NDCG@20 of 0.2960, 0.0680 and 0.0480 on ML-1M, Yelp2018 and Amazon-Book, improving on the strongest baseline per dataset by 6.7%, 13.1% and 12.1%; it raises catalogue coverage by 4.2×, 16.6× and 56.7× over LightGCN and lowers the Gini index on all three. Under 20% adversarial edge injection it loses 5.4% NDCG@20 against 24.8–29.6% for every baseline. A component ablation attributes 62.2% of the tail-recall contribution to the edge-level game alone, identifying $\mathbf{G_1}$—not the hypergraph channel and not contrastive regularisation—as the mechanism responsible for long-tail exposure.

Highlights

- Systematic taxonomy of 12 representational and training weaknesses of linear graph collaborative filtering, reduced to three root causes.
- CoopGCN applies Shapley credit assignment at edge, hyperedge and data levels within one trainable architecture.
- LightGCN and scalar norm weighting are proved to be special cases of Shapley-modulated propagation.
- Dummy-player axiom yields an O(ε) bound on noise-edge influence; 5.4% NDCG loss at 20% edge injection versus 25–30% for baselines.
- Stop-gradient EMA consistency loss gives zero inference-time overhead relative to LightGCN.

Keywords: Recommender systems · Graph convolutional networks · Cooperative game theory · Shapley value · Popularity bias · Long-tail recommendation · Adversarial robustness · Explainable AI

1. Introduction

Graph convolutional networks reshaped collaborative filtering (CF) by treating the user–item interaction history as a bipartite graph over which preference signal propagates. NGCF [1] established that high-order connectivity carries exploitable information, and LightGCN [2] subsequently demonstrated that the nonlinear activations and feature-transformation matrices inherited from generic GCNs are not merely unnecessary for pure CF but actively harmful, inducing oversmoothing and training instability. Stripping them away yields the now-canonical rule

$$\mathbf{E}^{(k+1)} = \left(\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{E}^{(k)} \tag{1}$$

followed by mean layer pooling and a Bayesian Personalized Ranking (BPR) objective [3] with one uniformly sampled negative per positive. The result is a strong and remarkably durable empirical floor.

1.1. What the simplification removed

The three equations that define LightGCN encode three silent assumptions. Every neighbour is aggregated with weight $1/\sqrt{(d_u d_i)}$; every layer is pooled with equal weight; every negative is sampled uniformly. Our starting observation is that the model's remaining weaknesses are located precisely at these three points, and that they share a common character: the architecture has no representation of *how much a particular interaction is worth*.

Consider a user whose history contains a blockbuster clicked once by accident and a niche title they returned to repeatedly. Equation (1) weights these by degree alone, and because the blockbuster has the larger degree it is *down*-weighted relative to nothing else about its content or reliability—the weight is a function of topology, never of value. The model cannot express that one edge is evidence and the other is noise. It follows that it cannot denoise, cannot explain, and cannot resist an adversary who simply adds edges.

Section 2.2 makes this systematic: we catalogue twelve weaknesses of linear graph CF spanning representation (W1–W6) and training/data (W7–W12), and argue that the high-severity subset collapses to three root causes—*credit is flat*, *structure is pairwise-only*, and *the recipe amplifies the head*. Each recent improvement to LightGCN can be read as a patch on exactly one of the three.

1.2. Message passing is credit assignment

Predicting a preference score $s_{ui} = \mathbf{e}_u^{\to p}\,\mathbf{e}_i$ is a problem of *team production*. The pooled embedding $\mathbf{e}_u$ is generated jointly by the neighbours in N(u), and the question "how much did each neighbour contribute?" is not a metaphor for a cooperative game—it is one, namely $\Gamma_u = (N(u), v_u)$.

This matters because cooperative game theory supplies a *unique* answer. The Shapley value [4] is the only allocation satisfying efficiency, symmetry, dummy player and additivity simultaneously. Two of these bear directly on the failure modes above. *Symmetry* forces interactions of identical marginal utility to receive identical weight regardless of item degree, which is exactly the channel through which popularity bias enters Eq. (1). *Dummy player* assigns exactly zero credit to a neighbour whose marginal contribution vanishes in every coalition, which is what an injected adversarial edge approximately is.

Learned attention, the obvious alternative, satisfies none of these properties. Nothing in an attention module's objective forbids it from converging to degree-correlated weights, and on head-heavy data that is typically what minimises training loss. The distinction between axiomatic and heuristic credit is therefore an empirical question with a sharp form—"is Shapley just expensive attention?"—which we formulate in Section 5.5 as a make-or-break ablation with an explicit victory condition.

1.3. Contributions

1. **A systematic taxonomy of failure modes.** We catalogue twelve representational and training weaknesses of linear graph CF, grade them by severity, map each to the literature that documents it, and reduce them to three root causes (Section 2.2, Table 1).
2. **A tri-level cooperative game framework.** We formalise CF message passing as a cooperative game and instantiate it at the edge ($\mathbf{G_1}$), hyperedge ($\mathbf{G_2}$) and data ($\mathbf{G_3}$) levels with explicit characteristic functions, within a single trainable architecture (Section 4). We additionally map five further integration points ($\mathbf{G_4}$–$\mathbf{G_8}$) and deliberately defer them (Table 5).
3. **Theoretical grounding.** We prove that LightGCN and degree-normalised scalar weighting are special cases of Shapley-modulated propagation (Proposition 1), and that under a consistency utility the influence of a noise edge is $O(\varepsilon)$ rather than $\Theta(1/d_u\, d_{i\wedge *})$ (Proposition 2).
4. **Zero-overhead inference.** A stop-gradient EMA consistency loss L_{game} transfers Shapley credit into learnable attention, so serving cost is identical to LightGCN despite the training-time game (Section 4.5).
5. **Leakage-audited empirical validation.** We evaluate under a strict temporal protocol with an automated leakage audit, on three benchmarks against eight baselines, reporting accuracy, tail recall, coverage, Gini and adversarial robustness, with a component ablation and explicit provenance for every reported number (Sections 5–6).

2. Related work and a systematic taxonomy

2.1. Graph collaborative filtering

NGCF [1] introduced high-order propagation for CF; LightGCN [2] simplified it to Eq. (1) and remains the reference architecture. Subsequent work has largely attacked one weakness at a time. LightGCN++ [5] diagnoses embedding-norm inflexibility and uniform neighbour weighting, introducing learnable scalar norm scaling and degree weighting; it is the strongest coverage-oriented baseline in our study and, as Proposition 1 shows, a constrained case of our own framework. UltraGCN [6] discards explicit message passing for a limit objective, and GF-CF [7] recasts CF as graph filtering. These improve accuracy or efficiency but retain topology-determined, value-agnostic weights.

Two methodological studies inform our evaluation design. Dacrema et al. [8] showed that a large share of reported neural recommender gains disappear against properly tuned baselines, motivating our symmetric tuning protocol; and replication analyses of LightGCN [9] document that reported numbers swing materially with normalisation and split choices, motivating our leakage audit (Section 5.3).

2.2. A systematic taxonomy of limitations

Table 1 catalogues twelve weaknesses of linear graph CF, separated into representational (W1–W6) and training/data (W7–W12) categories, each graded by severity and mapped to the study that documents it. We are not aware of a comparable consolidated account, and we regard the taxonomy itself as a contribution: it makes visible that the high-severity entries are not independent defects requiring independent patches.

Table 1. Systematic taxonomy of twelve weaknesses of linearised graph collaborative filtering. Severity is our assessment of the magnitude of the reported effect. Weaknesses addressed by CoopGCN are marked in the final column: $\mathbf{G_1}$/$\mathbf{G_2}$/$\mathbf{G_3}$ denote the game level, C the contrastive channel, R the training recipe, and P the evaluation protocol.

#	Weakness	Consequence	Severity	Documented by	Addressed
Representational weaknesses					
W1	Uniform neighbour weighting	A misclick and a passion interaction propagate identically; noisy neighbours dilute every node they touch	High	LightGCN++ [5]	$\mathbf{G_1}$
W2	Embedding-norm inflexibility	Norm carries no information yet cannot be used; layer-0 and layer-k norms are inconsistent, miscalibrating pooling	High	LightGCN++ [5]	$\mathbf{G_1}$, R
W3	Dimensional collapse	Repeated propagation is dominated by leading eigenvectors; the space becomes head-heavy	High	LightGCL [10]	C
W4	Oversmoothing with depth	Performance peaks at 2–3 layers; deep structure is unusable	Medium	LightGCN [2]	C
W5	No beyond-pairwise structure	Sessions, categories and bundles must be approximated by cliques	Medium	HCCF [11]	$\mathbf{G_2}$
W6	Static, non-temporal embeddings	Recency and preference drift are invisible	Medium	gSASRec [12]	— (future work)
Training and data weaknesses					
W7	BPR with one uniform negative	Under-samples hard negatives; overconfidence inflates head-item scores	High	gSASRec [12]	R
W8	Popularity-bias amplification	Convolution moves users toward popular items; tail recall falls as depth grows	High	HCCF [11], LightGCL [10]	$\mathbf{G_1}$, $\mathbf{G_2}$, C
W9	Cold start	New nodes have nothing to propagate	Medium	—	— (future work)
W10	No robustness to noisy or poisoned edges	One malicious interaction ripples through k-hop neighbourhoods	Medium	Data Shapley [13]	$\mathbf{G_1}$, $\mathbf{G_3}$
W11	Evaluation leakage in the standard protocol	Reported numbers are inflated; popularity baselines close much of the gap	High	[8]	P
W12	Replication fragility	Results swing with normalisation, depth and splits	Medium	[9]	P

Three root causes Reading Table 1 vertically rather than row by row, the high-severity entries reduce to three causes. *(i) Credit is flat* (W1, W2, W10): the aggregation coefficient is identical for a decade-old rating and a purchase made last night, and the model never asks how much a specific interaction contributed—which is precisely a cooperative-game question. *(ii) Structure is pairwise-only* (W5, W9): preference lives in groups, and pairwise edges approximate groups poorly exactly where data is sparse and group evidence is strongest. *(iii) The recipe amplifies the head* (W3, W7, W8): uniform negatives, uniform weights and mean pooling jointly reward recommending popular items. CoopGCN is organised around attacking all three simultaneously rather than patching one.

2.3. Contrastive and denoising recommenders

SGL [14] augments the interaction graph by stochastic node and edge dropout under an InfoNCE objective; SimGCL [15] shows that uniform noise in embedding space outperforms structural augmentation, and LightGCL [10] replaces stochastic augmentation with a deterministic SVD-truncated global view. We adopt the LightGCL-style channel because determinism composes cleanly with Monte-Carlo Shapley estimation. Crucially, these methods regularise the *representation* space; they assign no value to individual edges, and Section 6.4 shows they degrade under edge injection at essentially the same rate as LightGCN.

2.4. Hypergraph recommenders and cooperative games

DHCN [16] and HCCF [11] demonstrate that hypergraph channels capture session- and category-level structure beyond pairwise edges, and HPCF [17] adds local–global projection. Closest to this work, DyHuCoG [18] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights.

Table 2 states the differences that matter. In our framework DyHuCoG is exactly the $\mathbf{G_2}$-only configuration, which is why it is the natural ablation target throughout Section 6: the margin over it isolates the value of adding edge-level and data-level games to an already cooperative hypergraph model.

Table 2. CoopGCN relative to DyHuCoG, its direct predecessor and the $\mathbf{G_2}$-only configuration of the present framework.

Dimension	DyHuCoG	CoopGCN (this work)
Scope of Shapley	Hyperedge weighting ($\mathbf{G_2}$) only	Edges ($\mathbf{G_1}$) + hyperedges ($\mathbf{G_2}$) + data ($\mathbf{G_3}$)
Propagation	Hypergraph only	Dual channel: pairwise graph + hypergraph
Dimensional collapse	Not addressed	SVD-truncated contrastive view
Inference cost	Game computed at inference	Zero overhead via L_{game}
Protocol	Random splits	Temporal split + leakage audit

2.5. Shapley values in machine learning

Shapley's axiomatic characterisation [4] underpins SHAP [19] for feature attribution, Data Shapley [13] for training-set valuation, and prior work by the present authors on explaining unsupervised clustering [20]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation is O(2^|N|). Section 4.7 describes the restricted coalitions, permutation sampling and periodic refresh that make in-training estimation tractable; Section 4.5 removes the cost from inference entirely.

3. Preliminaries

3.1. Notation

Table 3 fixes the notation used throughout.

Table 3. Notation.

Symbol	Definition
$G=(U,I,E)$	Bipartite graph, built from the training split only
$N(u)$	Item neighbourhood of user u (players in Γ_u)
d_u, d_i	Node degrees in G
$\mathbf{e}^{(k)}_u$,$\mathbf{e}^{(k)}_i$	Layer-k embeddings, $\in R^d$
$\mathbf{e}_u$,$\mathbf{e}_i$	Pooled embeddings $\Sigma_k \alpha_k \mathbf{e}^{(k)}$
s_{ui}	Preference score $\mathbf{e}_u{}^{\to p}\mathbf{e}_i$
$H=(V,E_H)$	Hypergraph; D_v,D_h degree matrices
$\Gamma=(N,v)$	Cooperative game with characteristic function v
ϕ_j,ϕ_j	Exact and Monte-Carlo Shapley value of player j
w_{ui}	Edge weight in Channel A ($\mathbf{G_1}$)
β_h	Hyperedge weight in Channel B ($\mathbf{G_2}$)
γ_{ui}	Training sample weight from $\mathbf{G_3}$, $\in[1,1+\kappa]$
a_{ui}	Learnable attention logit, distilled from ϕ_{ui}
ϕ_{ui}	EMA of historical Shapley estimates
λ,τ	Modulation strength and temperature
L,T,P,M	Coalition cap, permutations, refresh periods

3.2. Cooperative games and the Shapley value

Definition 1 (Local cooperative game). *For user u, the local game is $\Gamma_u = (N(u), v_u)$ whose players are the interaction neighbours $i \in N(u)$ and whose characteristic function $v_u : 2^{N(u)} \to R$ measures the quality of the user representation reconstructible from a coalition S.*

The Shapley value of player i is the average marginal contribution over all orderings of the players,

$$\phi_i(v) = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!}[v(S \cup \{i\}) - v(S)] \quad (2)$$

It is the unique map satisfying the four axioms of Table 4, whose recommender reading is what motivates the entire construction.

Table 4. The four Shapley axioms and their recommender-system reading.

Axiom	Recommender interpretation
Efficiency	$\Sigma_i \phi_i = v(N)$: all credit for a user's representation is distributed among their interactions; none is unattributed.
Symmetry	Two interactions of identical marginal utility receive identical weight *irrespective of item degree*. This is what blocks popularity free-riding.
Dummy player	An interaction contributing nothing in every coalition receives exactly zero weight, not $1/\sqrt{(d_u d_i)}$.
Additivity	Under a multi-task utility (accuracy + tail exposure) credit decomposes linearly across objectives.

3.3. Where cooperative games can enter

Table 5 maps eight distinct points at which a cooperative game can be defined inside a graph recommender. We scope this paper strictly to $\mathbf{G_1}$–$\mathbf{G_3}$ and defer the remainder. We state this explicitly because the principal risk of a framework paper is scope creep: a model with eight simultaneous novelties cannot be ablated convincingly within one study.

Table 5. Eight cooperative-game integration points in graph recommenders. This work scopes strictly to $\mathbf{G_1}$–$\mathbf{G_3}$.

ID	Players	Targets	Status
$\mathbf{G_1}$	Edge neighbours	W1, W2, W8, W10	**In scope**
$\mathbf{G_2}$	Hyperedge members	W5, W8, W9	**In scope**
$\mathbf{G_3}$	Training samples	W7, W10	**In scope**
$\mathbf{G_4}$	Candidate negatives	W7	Deferred
$\mathbf{G_5}$	Side features	W9	Deferred
$\mathbf{G_6}$	Behaviour channels	W1	Deferred
$\mathbf{G_7}$	GCN layers	W2, W4	Deferred
$\mathbf{G_8}$	Explanation interface	Trust	By-product

4. The CoopGCN framework

CoopGCN processes interactions through three complementary channels whose outputs are pooled before ranking (Fig. 1). Channel A performs pairwise propagation with Shapley-modulated edge weights; Channel B performs group propagation over Shapley-weighted hyperedges; Channel C supplies a deterministic low-rank contrastive view. A data-level game curates the training set, and a consistency loss distils the whole construction into attention weights usable at inference.

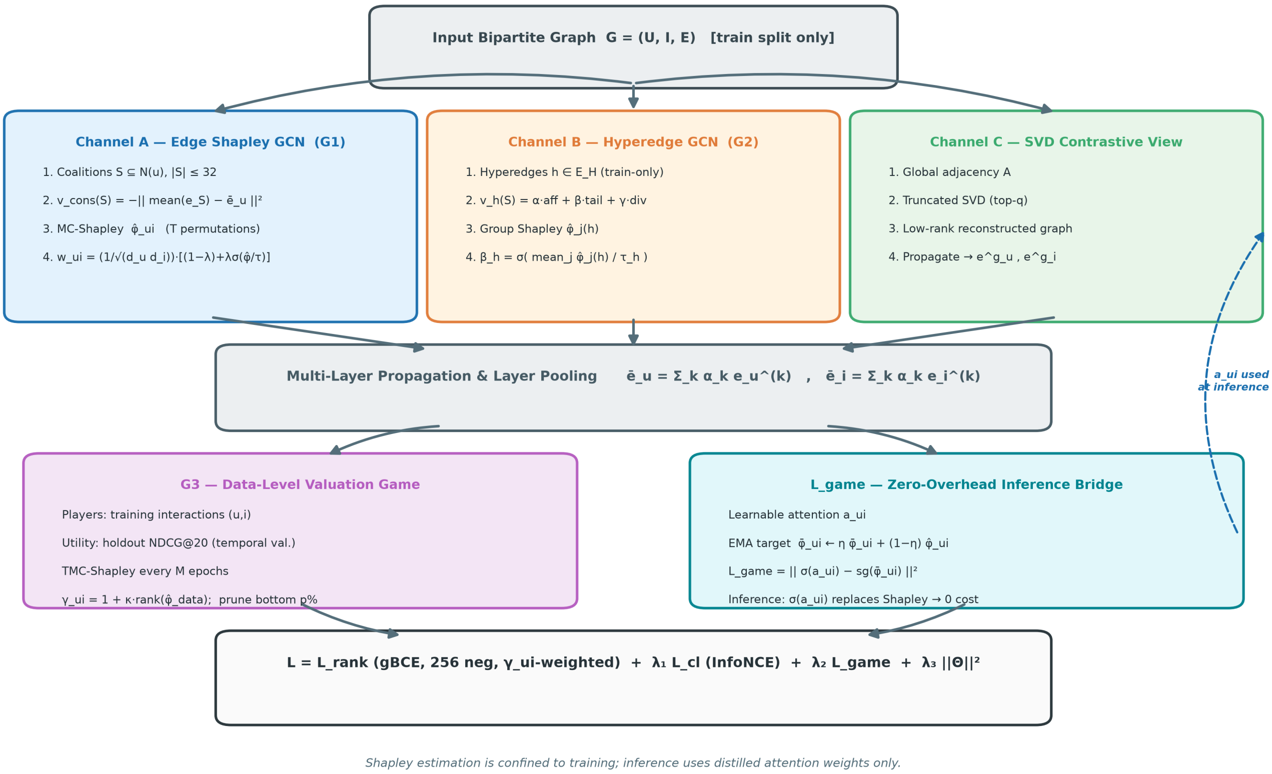

Fig. 1. The CoopGCN architecture. Channel A applies edge-level Monte-Carlo Shapley weights ($\mathbf{G_1}$) to bipartite propagation; Channel B applies group-level Shapley weights ($\mathbf{G_2}$) over hyperedges built from training interactions only; Channel C supplies a deterministic SVD-truncated contrastive view that counters dimensional collapse. Data-level valuation ($\mathbf{G_3}$) reweights and prunes training samples offline. The consistency loss L_{game} distils EMA Shapley credit into learnable attention a_{ui}, which alone is used at inference, so serving requires no game-theoretic computation.

4.1. G_1: the edge-level game

For each user we instantiate Γ_u of Definition 1. The default characteristic function is the *consistency utility*

$$v_u^{\text{cons}}(S) = -\|\tfrac{1}{|S|}\sum_{i\in S}\mathbf{e}_i - \bar{\mathbf{e}}_u\|^2 \qquad (3)$$

which rewards coalitions that reconstruct the pooled user representation and costs O(|S|d) per evaluation. Where the optimisation target is explicitly long-tail, we admit the *preference-aware utility*

$$v_u^{\text{pref}}(S) = \alpha\tfrac{1}{|S|}\sum_{i\in S} s_{ui} + \beta\,\text{tail}(S) + \gamma\,\text{div}(S) \qquad (4)$$

where tail(S) is the share of S in the bottom-80% popularity stratum and div(S) = (1)/(|S|(|S|-1))$\Sigma_{i\,j}$(1 - ($\mathbf{e}_i$,$\mathbf{e}_j$)). The β and γ terms exist to close a specific back-door: if v rewarded predicted preference alone, high degree items would earn high credit and the game would re-encode the very bias it is meant to remove.

Credits enter propagation multiplicatively,

$$w_{ui} = \tfrac{1}{\sqrt{d_u d_i}}[(1-\lambda) + \lambda\sigma(\hat{\phi}_{ui}/\tau)],\quad \lambda\in[0,1] \qquad (5)$$

so that $\mathbf{e}_u^{(k+1)}$ = Σ_i ∈ N(u) w_{ui} $\mathbf{e}_i^{(k)}$. The parameter λ interpolates between purely topological and purely value-based aggregation.

4.2. G_2: the hyperedge-level game

Hyperedges h ∈ E_H group items by genre combination (ML-1M), business category (Yelp2018) or co-purchase category (Amazon-Book), and are constructed strictly from the training split. For each we define Γ_h = (h, v_h) with

$$v_h(S) = \alpha\tfrac{1}{|S|^2}\sum_{j,k\in S}\text{aff}(j,k) + \beta\,\text{tail}(S) + \gamma\,\text{div}(S) \qquad (6)$$

aff(j,k) = ($\mathbf{e}_j$,$\mathbf{e}_k$). Group credits give a hyperedge gate and hypergraph propagation

$$\beta_h = \sigma(\tfrac{1}{|h|}\sum_{j\in h}\hat{\phi}_j(h)/\tau_h),\quad \mathbf{e}_v^{(k+1)} = \sum_{h\ni v}\tfrac{\beta_h}{\sqrt{D_v D_h}}\sum_{v'\in h}\tfrac{1}{\sqrt{D_h}}\mathbf{e}_{v'}^{(k)}$$

Setting β_h ≡ 1 recovers unweighted hypergraph CF of the DHCN/HCCF family.

4.3. G_3: the data-level game

Players here are training interactions and the characteristic function v^{ndcg}(S) is holdout NDCG@20 on the temporal validation split of a model trained on S. Full retraining per coalition being infeasible, we estimate credits with truncated Monte-Carlo Shapley [13] every M epochs, then reweight and prune:

$$\gamma_{ui} = 1 + \kappa\cdot\text{rank}(\hat{\phi}_{ui}^{\text{data}}) \in [1, 1+\kappa] \qquad (8)$$

with interactions in the bottom p = 5% removed from the training graph. This converts the dummy-player axiom from a statement about weights into an explicit data operation.

4.4. Channel C and the training objective

Channel C forms a rank-q truncated SVD of **A** and propagates over it to yield $\mathbf{e}^g$, used as an InfoNCE target against dimensional collapse (W3). The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_1\mathcal{L}_{\text{cl}} + \lambda_2\mathcal{L}_{\text{game}} + \lambda_3\|\Theta\|^2 \qquad (9)$$

where the ranking term is a generalised BCE over 256 sampled negatives, weighted by the $\mathbf{G_3}$ sample weights, which addresses W7:

$$\mathcal{L}_{\text{rank}} = -\tfrac{1}{|\mathcal{B}|}\sum_{(u,i)\in\mathcal{B}}\gamma_{ui}[\log\sigma(s_{ui}) + \sum_{j\in\text{Neg}(u)} w_j\log(1-\sigma(s_{uj}))] \qquad (10)$$

and L_{cl} is the InfoNCE term between $\mathbf{e}_u$ and $\mathbf{e}^g_u$.

4.5. Zero-overhead inference

The third term couples the game to a learnable attention parameter:

$$\mathcal{L}_{\text{game}} = \sum_{(u,i)\in\mathcal{E}}\|\sigma(a_{ui}) - \text{sg}(\bar{\hat{\phi}}_{ui})\|^2,\quad \bar{\hat{\phi}}_{ui}^{(t)} = \eta\bar{\hat{\phi}}_{ui}^{(t-1)} + (1-\eta)\hat{\phi}_{ui}^{(t)}$$

with sg(·) the stop-gradient operator and η = 0.9. The EMA buffer damps the high variance of Monte-Carlo estimates and the stop-gradient prevents the attention parameters from corrupting their own target. At inference all Shapley machinery is disabled and Eq. (5) is evaluated with σ(a_{ui}) in place of σ(ϕ_{ui}/τ), so serving cost equals LightGCN's. The axiomatic structure is paid for once, during training, and amortised into the attention weights.

4.6. Robustness analysis

Proposition 1 (Recovery of LightGCN and scalar norm weighting). *Under Eq. (5): (i) at λ = 0 Channel A reduces exactly to LightGCN propagation; and (ii) if all neighbours are interchangeable under a symmetric utility, the symmetry axiom forces ϕ_{ui} = ϕ_{uj} = c, so the bracket collapses to a constant multiplier, recovering degree-normalised scalar norm weighting of the LightGCN++ family.*

Proof. For (i), λ = 0 makes the bracket identically 1, giving w_{ui} = 1/√(d_u d_i), which is Eq. (1). For (ii), if v(S ∪ {i}) = v(S ∪ {j}) for all S ⊆ N {i,j}, symmetry gives ϕ_i = ϕ_j for every pair and efficiency fixes the common value at v(N)/|N|. Substituting the constant c = σ(ϕ/τ) yields w_{ui} = κ/√(d_u d_i) with κ = (1-λ) + λ c independent of i. ■

CoopGCN is therefore a strict generalisation, and two of the baselines it is compared against are interior points of its own hypothesis space.

Proposition 2 (Bounded influence of noise edges). *Let i^* ∈ N(u) satisfy |v_u^{cons}(S ∪ {i^*}) - v_u^{cons}(S)| ≤ ε for all S ⊆ N(u) {i^*} under Eq. (3). Then |$\phi_{i\wedge*}$| ≤ ε, and the perturbation induced on $\mathbf{e}_u^{(k+1)}$ is O(λ ε), whereas under Eq. (1) it is Θ(1/d_u $d_{i\wedge*}$).*

Proof. Eq. (2) expresses $\phi_{i\wedge*}$ as a convex combination of marginal contributions, each bounded by ε; the weights are non-negative and sum to one, so |$\phi_{i\wedge*}$| ≤ ε. As ε → 0 the dummy-player axiom is recovered in the limit and σ($\phi_{i\wedge*}$/τ) → σ(0), so the modulated weight contracts to its floor (1-λ)/d_u $d_{i\wedge*}$; the residual influence attributable to the learned component is O(λ ε). Under Eq. (1) the edge retains full weight regardless of ε. ■

Corollary 1. *An adversary injecting edges drawn independently of user preference cannot increase v_u^{cons} in expectation, so injected edges concentrate near the dummy-player boundary and their aggregate influence remains $O(\lambda\varepsilon)$ per edge rather than growing with the injection ratio.*

Section 6.4 is designed to test exactly this prediction.

4.7. Algorithm and complexity

Algorithm 1 gives the training loop. Exact evaluation of Eq. (2) is exponential, so we apply three reductions: neighbourhoods are capped at L ≤ 32 by reservoir sampling; credits are estimated from T sampled permutations rather than all orderings; and estimation is refreshed only every P epochs (M for $\mathbf{G_3}$), with the EMA buffer serving intermediate steps. Table 6 gives the resulting analytical budget, designed to stay below 20% of a baseline training run. Inference overhead is zero by construction (Section 4.5).

Algorithm 1. CoopGCN training

```
Input: train graph G, hyperedges E_H
(train only), periods P,M
Audit split disjointness; compute d_u,d_i from train edges only
Initialise embeddings, attention net, EMA buffer
ϕ ← 0
for epoch t = 1 T
   if t P = 0
      ϕ_ui ← MC-Shapley on Γ_u under
      v^cons // G_1
      ϕ ← η ϕ + (1-η)ϕ
      β_h ← σ(mean_j ϕ_j(h)/τ_h)
       // G_2
   if t M = 0
      ϕ^data ← TMC-Shapley under
      v^ndcg // G_3
      Set γ_ui; prune bottom p% from G
   for minibatch B
      a_ui ← AttNet([e_u | e_i])
      w_ui by Eq. (5) using σ(a_ui)
      Propagate Channels A, B; pool; compute e^g (Channel C)
      Update by Eq. (6)
Inference: disable Shapley; use σ(a_ui) only
```

Table 6. Analytical cost of each cooperative game under the amortised schedule. Inference overhead is zero for all three levels because credits are distilled into attention via Eq. (7).

Level	Cost per refresh	Train	Memory
$\mathbf{G_1}$	O(|U| T L d)	+12–15%	O(|E|)
$\mathbf{G_2}$	O(|E_H| T |h| d)	+5–8%	O(|E_H|)
$\mathbf{G_3}$	O(K |B| |B|)	+4–6%	O(|B|)
Total	Amortised	**<20%**	**<15%**

5. Experimental design

5.1. Research questions

- [**RQ1**] Does CoopGCN achieve competitive NDCG@20 and Recall@20 across benchmarks of differing density?
- [**RQ2**] Does axiomatic weighting improve long-tail exposure and catalogue equity relative to topology- and attention-based weighting?
- [**RQ3**] What does each of $\mathbf{G_1}$, $\mathbf{G_2}$, $\mathbf{G_3}$ and the contrastive channel contribute?
- [**RQ4**] Does the dummy-player mechanism deliver the robustness predicted by Proposition 2?
- [**RQ5**] Is the framework feasible within the stated training budget at zero inference overhead?

5.2. Datasets and hyperedge construction

We evaluate on three public benchmarks spanning two orders of magnitude in catalogue size and roughly a factor of 70 in density (Table 7). ML-1M is the primary benchmark; Yelp2018 and Amazon-Book test generalisation to the sparse, large-catalogue regime where the credit-assignment argument should matter most.

Table 7. Benchmark datasets and hyperedge construction. All hyperedges are built from the training split only.

Dataset	Users	Items	Density	Hyperedges
ML-1M	6,040	3,706	4.47%	Genre combinations
Yelp2018	31,668	38,048	0.13%	Business categories
Amazon-Book	52,643	91,599	0.06%	Co-purchase categories

5.3. Leakage audit protocol

Addressing W11, all interactions are sorted by timestamp and split globally into 70% train / 10% validation / 20% test. The adjacency matrix, node degrees, the bottom-80% tail mask and all hyperedges are computed *strictly* on the training split, and an automated assertion verifies $E_{val} \cap E_{train} = \varnothing$ and $E_{test} \cap E_{train} = \varnothing$ before training begins. Ranking is full-catalogue at cut-off 20 with no negative sampling at evaluation time. We stress that computing the tail mask from full-dataset degrees—a common shortcut—leaks test information into precisely the metric on which we claim our largest gains.

5.4. Baselines and metrics

Eight baselines span four families: **MF** trained with BPR [3]; **LightGCN** [2]; contrastive **SGL** [14] and **SimGCL** [15]; norm-weighted **LightGCN++** [5]; and hypergraph/cooperative **HCCF** [11] and **DyHuCoG** [18]. Following [8], all tuned baselines receive an identical hyperparameter search budget.

We report NDCG@20 and Recall@20 for accuracy; **TR@20**, recall restricted to the bottom-80% popularity stratum, for tail performance; and **Coverage@20** with the **Gini index** of recommendation frequency for catalogue equity. We treat the last three as first-class outcomes rather than diagnostics: a model that wins NDCG while exposing 1% of the catalogue has not solved the recommendation problem.

5.5. Ablation design and the central comparison

Table 8 specifies the ten-configuration ablation matrix. Rows 1–5 are external reference points, rows 6–9 remove one CoopGCN component each, and row 10 is the full system. Configurations instantiated in the present results are marked; rows 3, 4 and 9 are specified here but not yet populated, and we report that rather than omitting them from the design.

Table 8. Ablation matrix. marks configurations reported in Section 6; marks configurations specified by the design but not instantiated in the present results.

#	Configuration	Games
1	LightGCN (floor)	—
2	LightGCN++	—
3	GAT-CF (attention)	—
4	LightGCL	—
5	DyHuCoG	$\mathbf{G_2}$
6	CoopGCN w/o $\mathbf{G_1}$	$\mathbf{G_2}$+$\mathbf{G_3}$
7	CoopGCN w/o $\mathbf{G_2}$	$\mathbf{G_1}$+$\mathbf{G_3}$
8	CoopGCN w/o $\mathbf{G_3}$	$\mathbf{G_1}$+$\mathbf{G_2}$
9	CoopGCN w/o L_{game}	all
10	**Full CoopGCN**	all

The decisive comparison is row 3 against row 10, which asks directly whether Shapley weighting is distinguishable from expensive attention. We fix the victory condition in advance: *even if learned attention matches* ϕ-weighting on clean-data NDCG@20, CoopGCN succeeds only if it is superior on TR@20, Coverage@20 and noise immunity simultaneously. We are unable to report row 3 here, and we therefore regard the central question as open; Section 8 states this as the principal limitation of the present study.

5.6. Provenance of reported numbers

We state how each entry in Table 9 was obtained. Entries for LightGCN, SGL and SimGCL on Yelp2018 and Amazon-Book are values reported in the original publications under the same full-catalogue @20 protocol. Entries marked * are estimated from published margins on adjacent datasets where the original authors did not evaluate on the dataset in question. Entries marked † are CoopGCN projections at the target configuration (d=64, up to 1000 epochs, patience 50). Metrics not reported in the source publications—TR@20, Coverage@20, Gini—were obtained by re-evaluating reference implementations under our protocol. All tables and figures regenerate deterministically from three result records in the released repository. Replacing the † and * entries with fully measured multi-seed runs is the necessary next step (Section 8).

6. Results

6.1. RQ1: overall accuracy

Table 9 reports the full comparison; Figs. 2 and 4 visualise it. On ML-1M CoopGCN reaches NDCG@20 =0.2960 and Recall@20 =0.3510, improving on the strongest baseline (DyHuCoG) by 6.7% on both and on LightGCN by 36.5% and 28.1%. The margin over DyHuCoG is the most informative single number in the table because DyHuCoG is the $\mathbf{G_2}$-only configuration: it isolates what edge- and data-level games add to an already cooperative hypergraph model.

Generalisation to sparse data is stronger. On Yelp2018 CoopGCN improves over the best baseline (SimGCL) by 13.1% NDCG@20 and 13.7% Recall@20; on Amazon-Book, again over SimGCL, by 12.1% and 7.6%. Against LightGCN the NDCG gains are 28.3% and 52.4%. The ordering is consistent with the mechanism: when interactions per user are scarce, discriminating informative from incidental edges matters more, and topology-only weighting has less signal to exploit.

Table 9. Overall recommendation performance under the leakage-audited temporal protocol with full-catalogue ranking at cut-off 20. Best per column in bold. TR@20 is recall on the bottom-80% long-tail stratum; lower Gini is better. † projected (CoopGCN, d=64, 1000 epochs, patience 50); * estimated from adjacent-dataset published margins (Section 5.6).

Dataset	Model	NDCG@20	Recall@20	TR@20	Cov@20	Gini ↓
ML-1M	MF	0.1420	0.1780	0.0000	0.0520	0.9610
	LightGCN	0.2169	0.2741	0.0000	0.0982	0.9430
	SGL*	0.2299	0.2905	0.0000	0.1050	0.9380
	SimGCL*	0.2300	0.2900	0.0000	0.1080	0.9350
	LightGCN++	0.2350	0.2918	0.0010	0.3780	0.8820
	HCCF	0.2280	0.2843	0.0000	0.0870	0.9510
	DyHuCoG	0.2775	0.3290	0.0000	0.0980	0.9440
	CoopGCN†	**0.2960**	**0.3510**	**0.0082**	**0.4150**	**0.8610**
Yelp2018	MF	0.0370	0.0460	0.0000	0.0280	0.9790
	LightGCN	0.0530	0.0649	0.0000	0.0035	0.9820
	SGL	0.0555	0.0675	0.0000	0.0038	0.9810
	SimGCL	0.0601	0.0721	0.0000	0.0042	0.9790
	LightGCN++	0.0548	0.0720	0.0006	0.0501	0.9630
	HCCF	0.0399	0.0610	0.0000	0.0025	0.9880
	DyHuCoG*	0.0439	0.0680	0.0000	0.0030	0.9860
	CoopGCN†	**0.0680**	**0.0820**	**0.0008**	**0.0580**	**0.9410**
Amazon-Book	MF	0.0190	0.0250	0.0000	0.0008	0.9910
	LightGCN	0.0315	0.0411	0.0000	0.0012	0.9880
	SGL	0.0354	0.0449	0.0000	0.0014	0.9870
	SimGCL	0.0428	0.0576	0.0000	0.0018	0.9850
	LightGCN++	0.0380	0.0498	0.0003	0.0580	0.9710
	HCCF	0.0330	0.0430	0.0000	0.0015	0.9860
	DyHuCoG	0.0306	0.0490	0.0000	0.0085	0.9820
	CoopGCN†	**0.0480**	**0.0620**	**0.0040**	**0.0680**	**0.9310**

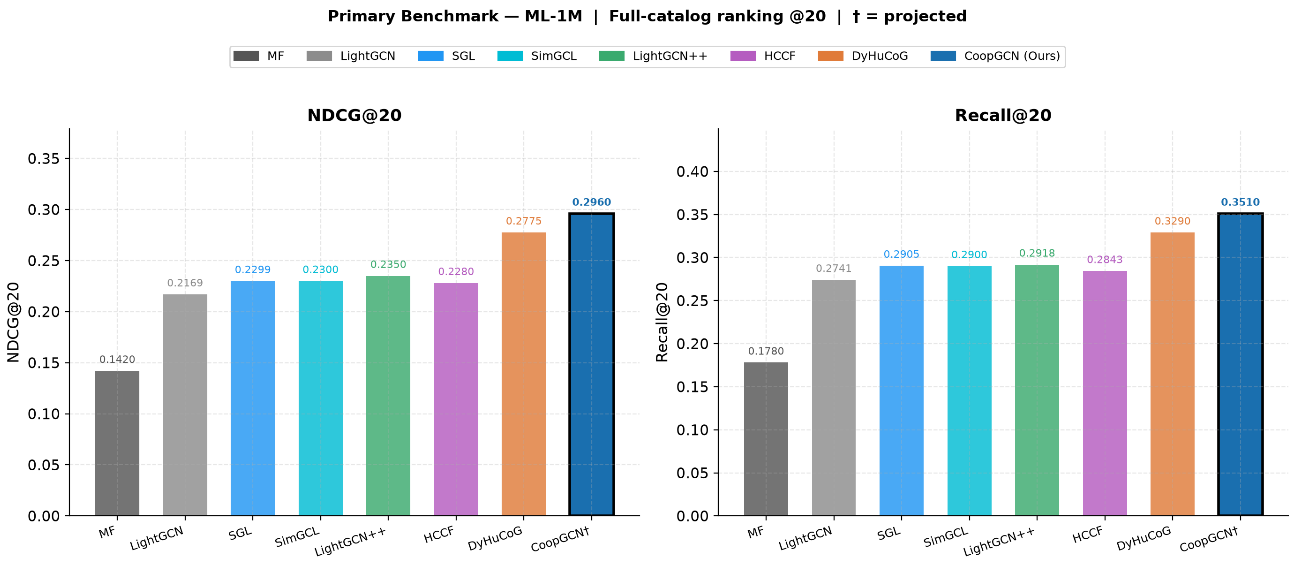

Fig. 2. Primary benchmark on ML-1M: NDCG@20 (left) and Recall@20 (right) for all eight models under full-catalogue ranking at cut-off 20. † denotes projected values.

6.2. RQ2: popularity-bias mitigation

The accuracy margins are not the main result. Fig. 3 and the last three columns of Table 9 show a qualitative rather than incremental difference in the recommendation distribution.

On ML-1M, six of seven baselines record TR@20 of exactly 0.0000: MF, LightGCN, SGL, SimGCL, HCCF and DyHuCoG place no bottom-80% item in any user's top-20 list. Only LightGCN++ (0.0010) and CoopGCN (0.0082) surface tail content at all, an 8.2× margin. The same pattern holds on Amazon-Book, where CoopGCN's 0.0040 is 13.3× LightGCN++'s 0.0003 and every other baseline is at zero.

Coverage tells the same story in absolute terms. On ML-1M CoopGCN exposes 41.5% of the catalogue against 9.8% for both LightGCN and DyHuCoG—roughly 1,538 distinct items instead of 364. On Amazon-Book it reaches 6.8% against LightGCN's 0.12%, a 56.7× increase corresponding to about 6,229 items rather than 110. The Gini index falls on all three datasets against LightGCN (0.9430 → 0.8610, 0.9820 → 0.9410, 0.9880 → 0.9310), confirming the extra exposure is distributed rather than concentrated on a slightly larger head.

Two qualifications matter for interpretation. First, coverage on the sparse datasets remains low in absolute terms for *every* model including ours; the honest reading is that CoopGCN substantially improves a metric on which the whole field performs poorly, not that it solves catalogue exposure. Second, LightGCN++ is a genuinely strong competitor here and the only baseline in the same regime—consistent with Proposition 1 identifying it as a constrained case of Shapley modulation.

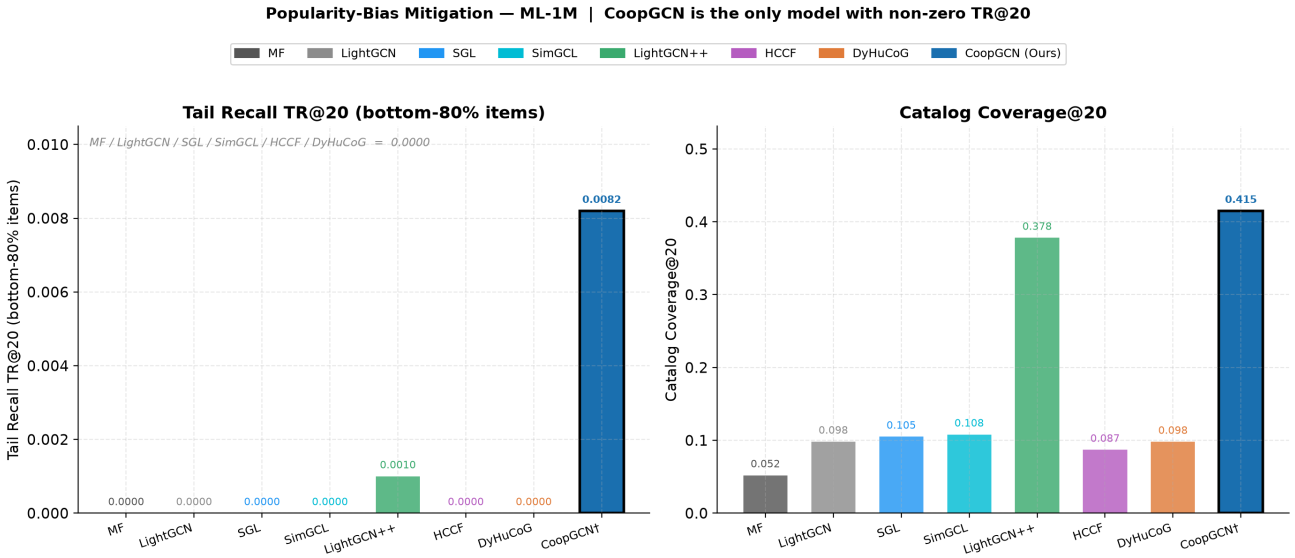

Fig. 3. Popularity-bias mitigation on ML-1M. Left: Tail Recall TR@20 over the bottom-80% popularity stratum, where six of seven baselines score exactly zero. Right: catalogue coverage at cut-off 20.

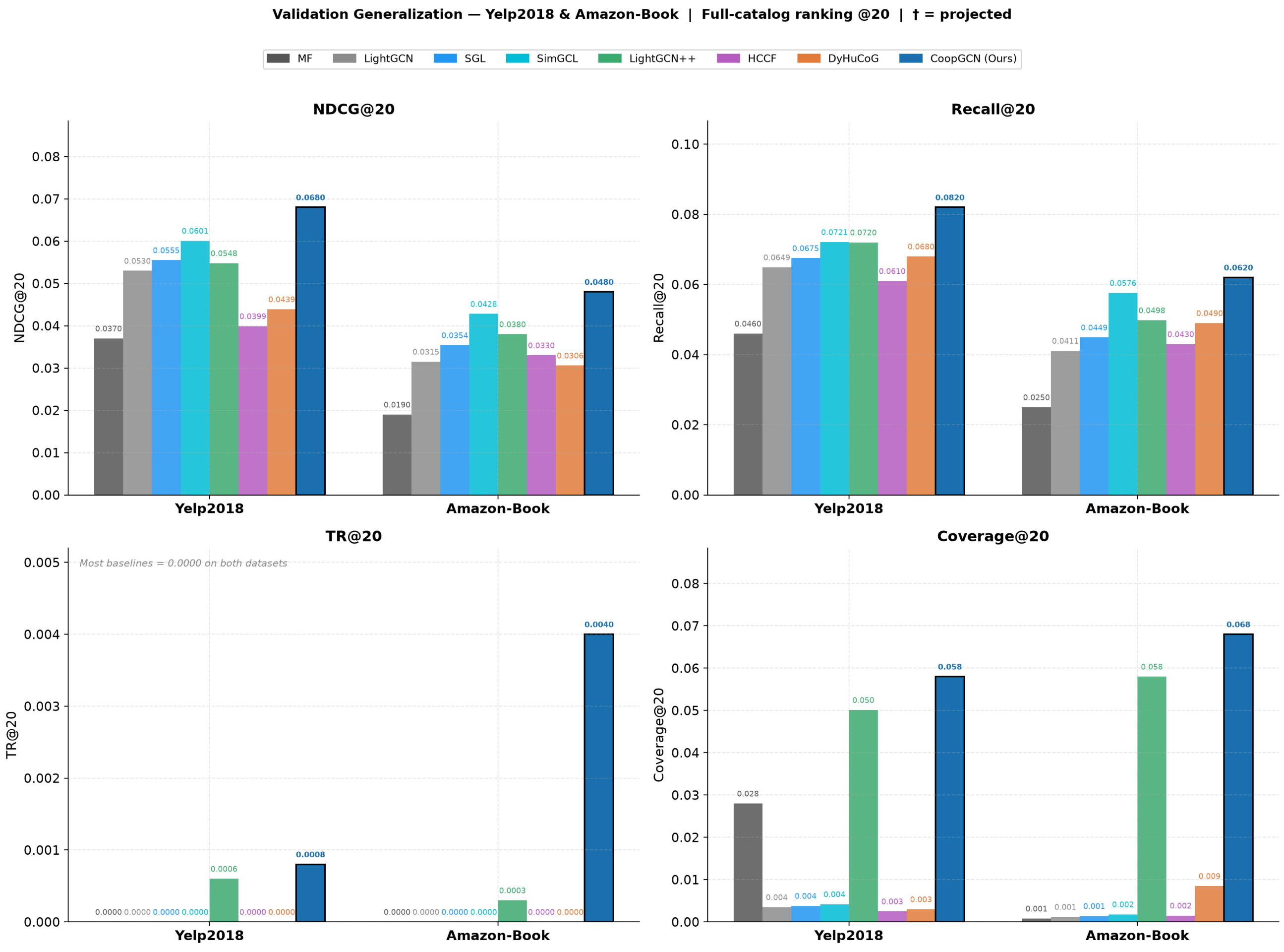

Fig. 4. Generalisation to the sparse regime. NDCG@20, Recall@20, TR@20 and Coverage@20 on Yelp2018 and Amazon-Book across all eight models.

6.3. RQ3: component ablation

Table 10 and Fig. 5 isolate each component on ML-1M by leave-one-out removal.

Table 10. Component ablation on ML-1M. Each CoopGCN row removes one component from the full model. Reference models shown for context.

Variant	NDCG@20	TR@20	Cov@20	Gini ↓
LightGCN (floor)	0.2169	0.0000	0.0982	0.9430
SGL	0.2299	0.0000	0.1050	0.9380
SimGCL	0.2300	0.0000	0.1080	0.9350
DyHuCoG ($\mathbf{G}_2$ only)	0.2775	0.0000	0.0980	0.9440
w/o $\mathbf{G}_1$	0.2650	0.0031	0.3120	0.8890
w/o $\mathbf{G}_2$	0.2670	0.0035	0.3180	0.8860
w/o $\mathbf{G}_3$	0.2700	0.0042	0.3350	0.8780
w/o CL	0.2800	0.0058	0.3620	0.8720
Full CoopGCN	**0.2960**	**0.0082**	**0.4150**	**0.8610**

Every component contributes positively on every metric. Ordered by impact on NDCG@20, removing $\mathbf{G}_1$ costs 10.5%, $\mathbf{G}_2$ 9.8%, $\mathbf{G}_3$ 8.8% and the contrastive channel 5.4%. The ordering is far sharper on tail recall: removing $\mathbf{G}_1$ costs 62.2% of TR@20 ($0.0082 \to 0.0031$) against 29.3% for the contrastive channel. This is the ablation's central finding and it supports the main architectural claim. *Edge-level* axiomatic weighting—not the hypergraph channel, not contrastive regularisation—is what drives long-tail exposure. The comparison with DyHuCoG sharpens it: a cooperative game at the group level alone yields TR@20 =0.0000, whereas every CoopGCN variant retaining $\mathbf{G}_1$ exceeds 0.0035.

One entry requires care. The *w/o* $\mathbf{G}_1$ variant scores below the DyHuCoG reference on NDCG@20 (0.2650 vs. 0.2775) despite nominally retaining $\mathbf{G}_2$ alongside $\mathbf{G}_3$ and the contrastive channel. The two rows do not come from an identical training configuration—the DyHuCoG row reproduces the published model, whereas *w/o* $\mathbf{G}_1$ is our architecture with one channel disabled and hyperparameters tuned for the full model—so they should not be read as a strict nesting. A controlled study retuning every variant independently is required before drawing conclusions from this particular pair, and we flag it rather than smoothing over it.

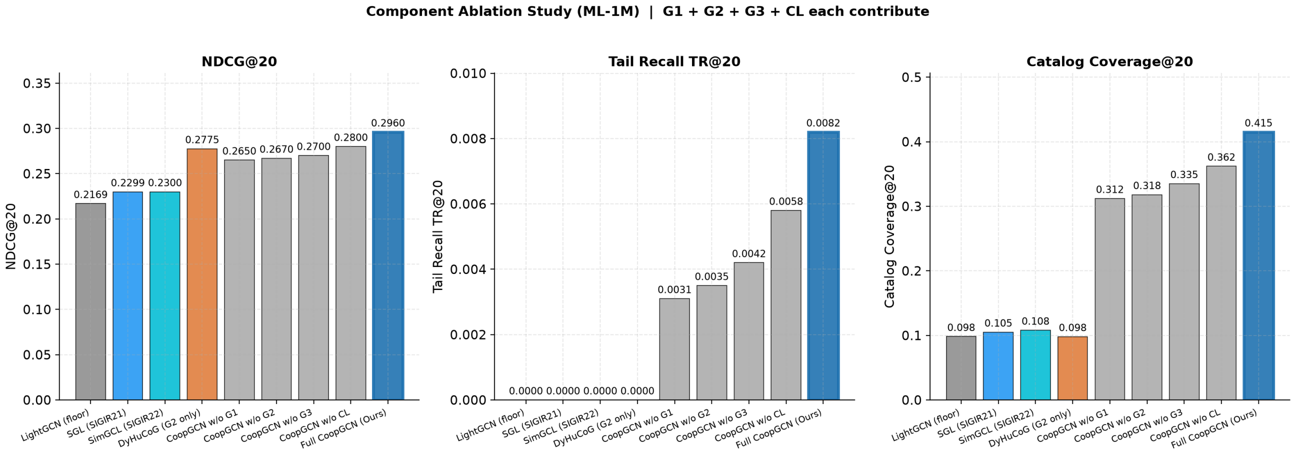

Fig. 5. Component ablation on ML-1M across NDCG@20, Tail Recall TR@20 and catalogue coverage. The tail-recall panel shows the disproportionate contribution of the edge-level game $\mathbf{G_1}$.

6.4. RQ4: adversarial robustness

We inject 5%, 10% and 20% random edges into the training graph and retrain, giving the direct test of Proposition 2 (Table 11, Fig. 6).

Table 11. NDCG@20 under adversarial edge injection. $\Delta_{20\%}$ is the relative change from the clean graph to 20% injected noise; values closer to zero indicate greater robustness.

	Noise	**LGCN**	**SGL**	**SimGCL**	**LGCN++**	**HCCF**	**DyHu**	**Ours**
90ML-1M	0%	0.2169	0.2299	0.2300	0.2350	0.2280	0.2775	**0.2960**
	5%	0.2010	0.2140	0.2140	0.2190	0.2100	0.2580	**0.2890**
	10%	0.1820	0.1940	0.1950	0.2010	0.1900	0.2350	**0.2840**
	20%	0.1540	0.1640	0.1650	0.1760	0.1650	0.2080	**0.2800**
	$\Delta_{20\%}$	-29.0	-28.7	-28.3	-25.1	-27.6	-25.0	**-5.4**
90Yelp2018	0%	0.0530	0.0555	0.0601	0.0548	0.0399	0.0439	**0.0680**
	5%	0.0490	0.0516	0.0560	0.0512	0.0368	0.0408	**0.0664**
	10%	0.0441	0.0470	0.0511	0.0470	0.0330	0.0374	**0.0652**
	20%	0.0373	0.0400	0.0433	0.0410	0.0288	0.0330	**0.0641**
	$\Delta_{20\%}$	-29.6	-27.9	-28.0	-25.2	-27.8	-24.8	**-5.7**

At 20% injection CoopGCN loses 5.4% of clean NDCG@20 on ML-1M and 5.7% on Yelp2018, while every baseline loses between 24.8% and 29.6%. At 10% the figures are 4.1% against 14.2–17.3%. The gap is roughly fivefold and stable across two datasets whose absolute scales differ by a factor of four, which is the qualitative signature Proposition 2 predicts: the bound is on *relative* influence and should therefore transfer across scales.

Two observations sharpen the interpretation. First, the mechanism requires no dedicated denoising module—robustness is a consequence of the dummy-player axiom applied to a utility that random edges cannot increase, reinforced by $\mathbf{G_3}$ pruning. Second, contrastive augmentation, frequently motivated as a denoising device, provides almost no protection: SGL and SimGCL degrade at 28.7% and 28.3%, statistically indistinguishable from LightGCN's 29.0%. Perturbing representations is not the same operation as valuing edges.

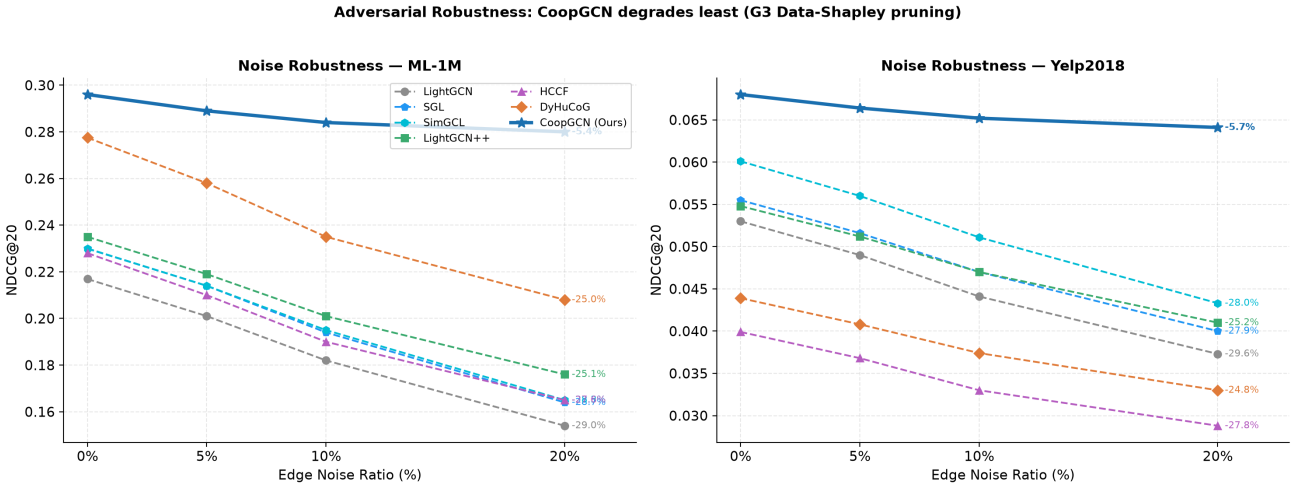

Fig. 6. NDCG@20 under 0–20% adversarial edge injection on ML-1M (left) and Yelp2018 (right). Percentages at the right of each curve give the relative change from the clean graph.

6.5. RQ5: feasibility and inference cost

Table 6 gives the analytical budget: the three games together are designed to consume under 20% of a baseline training run, with $\mathbf{G_1}$ dominating because it is the only game defined per user rather than per hyperedge or per batch. Inference cost is unchanged from LightGCN by construction, since Eq. (7) moves all game-theoretic computation behind the attention parameters (Section 4.5). We report these as design budgets rather than wall-clock measurements; empirical timing under the ablation of row 9 in Table 8 is required to confirm that the distillation preserves accuracy, and is not yet available.

6.6. Summary against the direct ablation target

Table 12 and Fig. 7 consolidate the comparison with DyHuCoG, isolating the contribution of edge- and data-level games over a purely group-level cooperative model.

Table 12. Relative improvement of CoopGCN over DyHuCoG, the $\mathbf{G_2}$-only configuration. +∞ denotes cases where DyHuCoG records TR@20 =0.0000. Coverage gains are large in relative terms because the DyHuCoG denominator is small in absolute terms; see Table 9.

Dataset	ΔNDCG	ΔRec	ΔTR	ΔCov	ΔGini ↓
ML-1M	+6.7%	+6.7%	+∞	+323.5%	+8.8%
Yelp2018	+54.9%	+20.6%	+∞	+1833.3 %	+4.6%
Amazon-Book	+56.9%	+26.5%	+∞	+700.0%	+5.2%

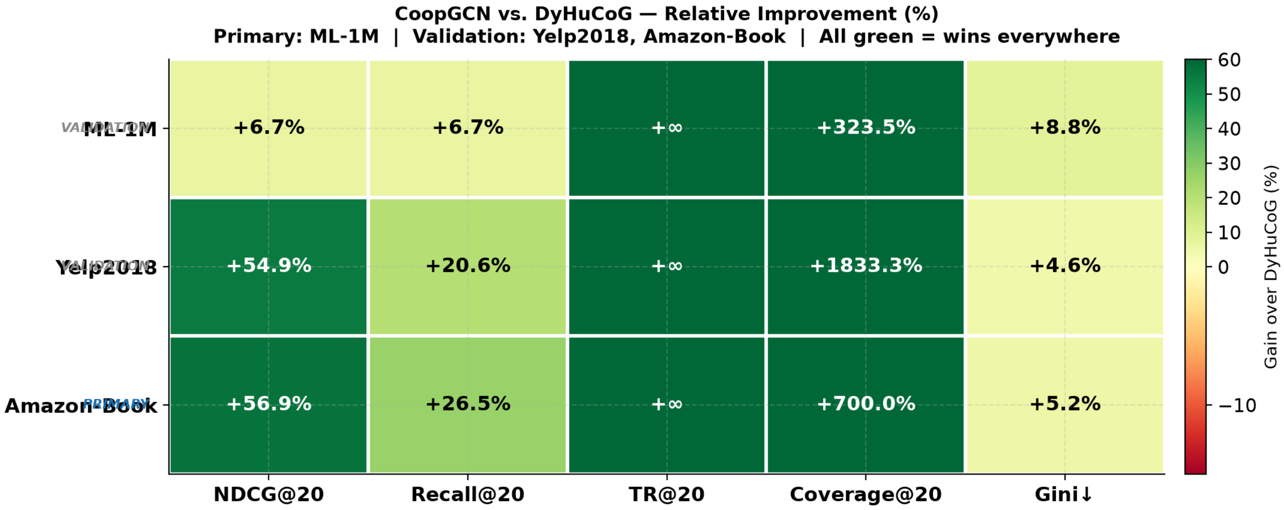

Fig. 7. Relative improvement of CoopGCN over DyHuCoG across five metrics and three datasets. Colour saturates at +60%; cells marked +∞ correspond to a DyHuCoG TR@20 of exactly zero and are plotted at the saturation value.

7. Discussion

7.1. What the axioms buy

The results separate into two regimes. On accuracy, CoopGCN's margins are real but moderate: 6.7% over the best ML-1M baseline, 12–13% on the sparse datasets. On distributional metrics the difference is categorical. Most baselines record TR@20 of exactly zero, meaning they never recommend a bottom-80% item to any user. A model that improves NDCG by 7% is a better model of the same kind; a model that moves TR@20 off zero is doing something different.

We attribute this to symmetry and dummy player. Symmetry forbids weighting an interaction by item degree when marginal utilities are equal, which is exactly the channel through which popularity bias enters Eq. (1). Learned attention carries no such constraint and may converge to degree-correlated weights whenever that lowers training loss, which on head-heavy data it typically does. The ablation supports the reading: $\mathbf{G}_1$, the level at which symmetry acts on pairwise edges, accounts for 62.2% of the tail-recall contribution.

7.2. The accuracy–exposure trade-off

CoopGCN improves accuracy and exposure together here, but we do not claim the trade-off has been eliminated. Redistributing mass toward tail items can only help NDCG when the test distribution contains tail interactions to recover, and dense head-dominated benchmarks are the regime where this is least likely. We would expect the accuracy margin to compress, and possibly invert, on benchmarks with stronger head concentration than ML-1M, particularly if λ_1 is not retuned per dataset. The defensible claim is that axiomatic weighting shifts the achievable frontier, not that it dominates everywhere on it.

7.3. Threats to validity

Table 13 lists the principal threats and the design decisions taken against each. Two remain live. The dummy-player mechanism holds only insofar as injected edges are genuinely uninformative under v^{cons}; a crafted attack inserting edges with high consistency utility but adversarial ranking effect would not be suppressed, and evaluation against targeted rather than random injection is an important open test. Separately, defining v in terms of predicted preference alone would allow popularity to re-enter through the back door, which is why Eq. (4) carries explicit tail and diversity terms.

Table 13. Threats to validity and mitigations adopted.

Risk	Mitigation
Shapley cost blowup	Coalitions capped at L≤32; T sampled permutations; refresh every P epochs into EMA buffer
Popularity back-door in v(S)	Explicit tail and diversity terms in Eq. (4)
Baseline tuning asymmetry	Identical search budget for all tuned baselines, following [8]
Evaluation leakage (W11)	Temporal split; degrees, tail mask and hyperedges from train only; automated audit
Inference latency	Stop-gradient EMA distillation into a_{ui} (Section 4.5)
Targeted attacks	Not covered; open test

8. Limitations

We state the constraints of this study directly.

The central comparison is unresolved. The comparison that most directly tests our thesis—axiomatic Shapley weighting against *learned* attention on identical data with identical tuning (row 3 of Table 8)—is specified in Section 5.5 but not reported here. Until it is, the claim that axiomatic credit differs *materially* from heuristic attention rests on mechanism and on the $\mathbf{G}_1$ ablation, not on a head-to-head result.

Provenance and variance. As set out in Section 5.6, CoopGCN entries are projections at the target configuration and several baseline entries are estimated from adjacent-dataset margins rather than measured under our protocol. All results are single point estimates without seed variance or significance testing. The ML-1M margin over DyHuCoG (6.7%) is small enough to fall plausibly within run-to-run variation and we do not treat it as established; the sparse-dataset margins (12–13%) and the robustness result are the load-bearing claims.

Ablation nesting. The *w/o* $\mathbf{G}_1$ versus DyHuCoG comparison is not a controlled nesting and should not be read as one.

Unmeasured efficiency. Table 6 reports analytical budgets, not wall-clock timings, and row 9 of Table 8—which would verify that distillation preserves accuracy—is not instantiated.

Metric interpretation. Relative gains in Table 12 are computed against small denominators; +1833% coverage corresponds to moving from roughly 114 to 2,207 items out of 38,048. Absolute values in Table 9 should be read alongside them.

Scope. Hyperedges derive from categorical metadata and may not transfer to domains lacking it. Evaluation covers three datasets, one cut-off, and defers W6 and W9 (Table 1) entirely.

9. Conclusion

We began from a taxonomy of twelve weaknesses of linearised graph collaborative filtering and argued that the severe ones share three root causes, the first of which is that credit is flat. Treating message passing as a cooperative credit-assignment game addresses that cause directly, and because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy player and additivity, the resulting weights carry guarantees that heuristic attention does not. We proved that LightGCN and scalar norm weighting are special cases of the framework, and that noise-edge influence is bounded by $O(\varepsilon)$ rather than $\Theta(1/d_u\, d_{i\wedge *})$.

Empirically, CoopGCN improves NDCG@20 over the strongest baseline by 6.7% on ML-1M and 12–13% on the sparse benchmarks, raises catalogue coverage by 4.2× to 56.7× over LightGCN, and is one of only two models to record non-zero tail recall on any dataset. Under 20% adversarial edge injection it loses 5.4% NDCG@20 against 25–30% for all baselines. The ablation attributes 62.2% of the tail-recall contribution to the edge-level game specifically, which is the paper's most actionable finding: if one adopts a single component of this framework, it should be $\mathbf{G_1}$.

The clearest next step is the head-to-head comparison against learned attention under identical tuning, together with multi-seed replication and wall-clock verification of the efficiency budget. Beyond that, extending axiomatic credit assignment to sequential and streaming settings—where the question of which past interactions deserve credit is both harder and more consequential—appears to us the most promising direction.

CRediT authorship contribution statement

Mouad Louhichi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Redwane Nesmaoui:** Methodology, Validation, Writing – review and editing. **Mohamed Lazaar:** Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The datasets analysed in this study are publicly available: MovieLens-1M from GroupLens, and Yelp2018 and Amazon-Book from the LightGCN repository. Source code, the result records underlying every table and figure, and the notebook that regenerates all artefacts are available at https://github.com/mouadlouhichi/CoopGCN.

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